MADAN MOHAN KUMAR

Full Stack Engineer – Java/Spring Boot/NodeJS| Bengaluru, Karnataka
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PROFESSIONAL SUMMARY

Full Stack Engineer with 3 years of experience delivering Java/Spring Boot, Node.js and ReactJS applications across REST APIs, JPA/Hibernate data layers, microservices and relational databases. Skilled in REST APIs, System design, application security and comfortable in SDLC, debugging, review, in Agile teams.

TECHNICAL SKILLS

Languages & Frameworks: Java, Spring Boot, Spring Security, JPA/Hibernate, Node.js, React.js, JavaScript

Databases: PostgreSQL (PostGIS), MySQL, MongoDB, schema design, indexing, query optimization

Cloud & DevOps: AWS (EC2, S3, SQS, IAM), Docker, Git/GitHub Actions (CI/CD)

Practices: Data Structures & Algorithms, Code Reviews, Debugging, Flyway Migrations

WORK EXPERIENCE

Software Engineer | Tata Consultancy Services (Johnson & Johnson engagement) | 3 YOE
Jul 2023 – Present

Engineered 20+ **role-based REST APIs** with JWT authentication and **RBAC** across 3+ internal applications serving 500 users.

- Desinged a Spring Boot assessment platform (User, Test, Question Bank, Test Attempt domains) with UUID primary keys and Flyway-managed schema migrations.
- Implementing stateless JWT auth via Spring Security, with query design that avoids N+1 issues in the JPA/Hibernate layer.
- Designed and optimized SQL and NoSQL schemas in PostgreSQL and MongoDB, improving indexing and query performance and cutting average query **response time by 20%** on large datasets.
- Built responsive React.js front ends with REST API integration, **reducing duplicate UI code** by 30% across 5 modules.
- Participated in Agile/Scrum teams, managing **sprint** tasks in Jira, conducting code reviews, debugging, deployment, and production support, while delivering 4+ product features.

PROJECTS

➢ **Remote Code Execution Engine** | *Spring Boot*, *WebSocket (STOMP/SockJS)*, *Docker*, *Redis*, K6 (Load Tested 10k+ users), p95

- Built a code-execution platform orchestrating **sandboxed Docker** containers via ProcessBuilder, with per-execution CPU/memory limits and real-time output streamed over Web Socket.
- Diagnosed a concurrency gap where the worker pool bounded queue throughput but not simultaneous container execution; implemented a **Semaphore**-based hard cap **on concurrent Docker** containers and verified correct permit hand-off via live Redis MONITOR tracing.
- Implemented a **Redis-backed job queue** to **throttle concurrent container spawns** and a multithreaded stream handler for stdout/stderr with timeout and cleanup handling.
- Managed per-session process lifecycle with **concurrent session registry**, resource cleanup, timeout handling, and kill-on-re-run.

➢ **Tutor School** | *React.js*, *Node.js*, *PostgreSQL/PostGIS*, *AWS*, *Nginx*

- Built secure REST APIs for Teacher/Parent/Admin dashboards with JWT auth and RBAC middleware, with zero unauthorized-access incidents during pilot.
- Implemented geospatial teacher search using PostGIS, achieving sub-200ms latency across 2,000+ profiles.
- Deployed to AWS EC2 behind Nginx with GitHub Actions CI/CD and CloudWatch monitoring.

EDUCATION

MCA – SASTRA Deemed University, Chennai | CGPA: 8.3 / 10

BCA – Amity University, Patna | CGPA: 8.5 / 10

CERTIFICATION

AWS Certified Developer – Associate (DVA-C02) | **Valid through Dec 2028**